



ROBOT EMBEDDED PROGRAMING

Lab 1: Review of Linux and Python for Robotics

Applications



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Objective

Introduce students to Linux, Python, and their importance in robotics, with a focus on building shared knowledge before starting technical work in later labs.

Session Outline

1. Welcome & Course Context (10 min)

- Overview of how the lab series fits into the course.
- Explanation of how Linux and Python connect to ROS 2 and real-world robotics.
- Short video or slides showing ROS 2 in action.

ROS uses the linux system as a low-cost driver device for connections via ethernet, USB or other connections. ROS 2 is the communication layer for robot instructions while we use Python for programming the logic into the ROS 2 system.

2. Linux in Robotics – Discussion (15 min)

- Instructor-led questions:
 - Where is Linux used in industry?
 - Why do many robotics systems use Linux instead of other operating systems?
- Show simple examples of Linux interfaces (terminal and GUI) on the projector.

Linux is used in Industry for all kinds of simple logic controllers or computers that will be going through repetitive tasks. Examples include PLC systems, cashier computers, safety systems and more. Linux is often used in the industry because it is a very customizable operating system that can be optimized for any task.

3. Python for Robotics – Discussion (15 min)

- Talk about Python's popularity and why it's chosen for ROS 2 development. - Mention common robotics-related libraries (e.g., rclpy, OpenCV, NumPy).
- Example: Python controlling a simulated robot (shown by instructor, not run by students).

Python is a popular choice for robot programming due to its ease of use, as well as its access to many useful libraries such as rclpy and NumPy, both of which we have used for robotics projects in the past.

4. Group Activity: Research & Whiteboard Sharing (30 min)

- Step 1: Students form small groups of 3–4.
- Step 2: Assign each group one of these topics:
 1. Advantages of Linux for robotics.
 2. Python's role in robotics.
 3. Key differences between ROS 1 and ROS 2.
 4. Examples of industries and projects using ROS 2.
- Step 3: Groups spend 15 minutes researching online or discussing based on prior knowledge.
- Step 4: Each group writes their 3–5 key points on the whiteboard.
- Step 5: Class discussion to compare findings.

5. Introduction to ROS 2 Jazzy Documentation (10 min)

- Navigate together to the official ROS 2 Jazzy docs:
<https://docs.ros.org/en/jazzy/index.html>
- Show where to find:
 - Tutorials
 - Concepts (Nodes, Topics, Services, Parameters)
 - Example projects
- Explain that no installation is needed yet—this will be done in Lab 2.

Deliverables

- Participation in group activity and class discussion.
- Students should leave with personal notes on:
 - Linux's importance in robotics.
 - Python's role and benefits.
 - One interesting fact learned from another group.

Preparation for Lab 2

- Review the “Beginner: CLI Tools” section of the ROS 2 Jazzy docs.
- Bring your laptop ready for installation activities next week.